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July 13, 2016

Docket Nos.: 50-366

NL-16-1090

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D. C. 20555-0001

Edwin I. Hatch Nuclear Plant
Licensee Event Report 2016-002-00
Group 1 Isolation Received During Turbine Testing

Ladies and Gentlemen:

In accordance with the requirements of 10 CFR 50.73(a)(2)(iv)(A) Southern Nuclear Operating Company hereby submits the enclosed Licensee Event Report.

This letter contains no NRC commitments. If you have any questions, please contact Greg Johnson at (912) 537-5874.

Respectfully submitted,

A handwritten signature in black ink, appearing to read 'David R. Vineyard', written over a horizontal line.

D. R. Vineyard
Vice President – Hatch

DRV/cdp

Enclosures: LER 2016-002-00

cc: Southern Nuclear Operating Company
Mr. S. E. Kuczynski, Chairman, President & CEO
Mr. D. G. Bost, Executive Vice President & Chief Nuclear Officer
Mr. D. R. Vineyard, Vice President – Hatch
Mr. M. D. Meier, Vice President – Regulatory Affairs
Mr. D. R. Madison, Vice President – Fleet Operations
Mr. B. J. Adams, Vice President – Engineering
Mr. G. L. Johnson, Regulatory Affairs Manager - Hatch
RTYPE: CHA02.004

U. S. Nuclear Regulatory Commission
Ms. C. Haney, Regional Administrator
Mr. M. D. Orenak, NRR Project Manager – Hatch
Mr. D. H. Hardage, Senior Resident Inspector – Hatch

Edwin I. Hatch Nuclear Plant Unit 2

LER 2016-002-00

Group 1 Isolation Received During Turbine Testing

**LICENSEE EVENT REPORT (LER)**
(See Page 2 for required number of digits/characters for each block)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by internet e-mail to Infocollections.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME Edwin I. Hatch Nuclear Plant Unit 2	2. DOCKET NUMBER 05000 366	3. PAGE 1 OF 3
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4. TITLE
Group I Isolation Received During Turbine Testing

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
05	23	2016	2016	- 002	- 00	07	14	2016	FACILITY NAME	DOCKET NUMBER

9. OPERATING MODE	11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)			
4	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. POWER LEVEL 0	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ OTHER	Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT Edwin I. Hatch / Carl James Collins – Licensing Supervisor	TELEPHONE NUMBER (Include Area Code) 912-537-2342
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13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX

14. SUPPLEMENTAL REPORT EXPECTED

☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO

15. EXPECTED SUBMISSION DATE

MONTH	DAY	YEAR

ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

On May 23, 2016, at 1009 EDT, while personnel were performing turbine testing with Unit 2 offline for planned maintenance, an event resulted in the actuation of containment isolation valves in more than one system. In response to this unexpected signal, 2B21-F016 (Steam Line Drain Line Inboard Isolation Valve), 2B21-F019 (Steam Line Drain Line Outboard Isolation Valve), and 2B31-F019 (Reactor Water Sample Inboard Isolation Valve) went closed, all of which are primary containment isolation valves actuated by Group I Isolation. The Group I Isolation signal initiated based on low condenser vacuum during the turbine testing procedure, a valid condition for actuation that was expected to have been bypassed in the logic during the performance of this procedure. Inadequate procedure usage caused these systems to actuate in a way that was not part of the planned evolution. Although the Unit was shut down when this signal was received, and primary containment isolation was not required to mitigate the consequences of an event, this isolation signal has been determined to have been valid due to the initiation in response to actual plant conditions or parameters which satisfy the requirements for initiation of the system

As a corrective action, the operating procedure will be revised to clarify the required conditions to perform Turbine Trip Testing.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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		YEAR	SEQUENTIAL NUMBER	REV NO.
Edwin I. Hatch Nuclear Plant Unit 2	05000-366	2016	- 002	- 00

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

General Electric- Boiling Water Reactor (BWR)
Energy Industry Identification System codes appear in the text as (EISS Code XX)

DESCRIPTION OF EVENT

On May 23, 2016, at 1009 EDT, with Unit 2 at zero percent rated thermal power, a full Group I Isolation signal was received during the performance of a turbine operability test [EISS Code TA; Main Turbine System]. In response to this unexpected signal, 2B21-F016 (Steam Line Drain Line Inboard Isolation Valve), 2B21-F019 (Steam Line Drain Line Outboard Isolation Valve), and 2B31-F019 (Reactor Water Sample Inboard Isolation Valve) [EISS Code TF; Turbine Drains and Miscellaneous Piping System] went closed.

The turbine testing evolution was stopped and investigation commenced to determine the cause of the Group I isolation. Investigation revealed the Condenser Low Vacuum trip was not bypassed as expected prior to the test performance per the Turbine Testing Trip Overrides and Valve Stroking procedure. There were no inoperable structures, systems, or components (SSCs) which contributed to this event.

CAUSE OF EVENT

The cause of the actuation of the Group I isolation signal during Turbine Testing was due to inadequate procedure usage. Poor communications between maintenance and operations personnel resulted in false assumptions that the Group I isolation trips were bypassed. Causal factors included procedure weaknesses which resulted in improper interpretation of critical steps.

CORRECTIVE ACTIONS

Immediately following the Group I isolation, Turbine Testing was halted. The operating procedure is being revised to clarify the required conditions to perform Turbine Trip Testing. Lessons learned were communicated among the Operations department. Additional training and qualification actions were taken to address the human performance aspects of this event.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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NARRATIVE**REPORTABILITY ANALYSIS AND SAFETY ASSESSMENT**

This event is reportable under 10 CFR 50.73(a)(2)(iv)(A) as an event or condition that resulted in automatic actuation of general containment isolation signals affecting containment isolation valves in more than one system or multiple main steam isolation valves (MSIVs). This isolation signal was valid in response to actual plant conditions induced by testing, and impacted isolation valves in the Nuclear Boiler System and the Reactor Recirculation System.

This turbine testing was being conducted with the reactor offline when the drywell was open for maintenance activities. The received Group I isolation signal was valid due to conditions measured by the appropriate instruments, however the isolation was not necessary to protect against an actual radiological event in progress. The main steam isolation valves were already closed prior to the start of the testing. Based on this information this reported condition had low safety consequence.

ADDITIONAL INFORMATION

Other Systems Affected: Isolation valves in the Nuclear Boiler System and the Reactor Recirculation System automatically went closed.

Failed Components Information: None.

Commitment Information: This report does not created any new licensing commitments.

Previous Similar Events:

LER 2009-005-0

On May 15, 2009 at 05:19 EDT, Unit I was in the cold shutdown mode with reactor power at 0 CMWT. At that time a Group I isolation signal on actual low condenser vacuum was generated when Operations personnel moved the reactor mode switch to the run position per subsection 7.1 of special purpose procedure 42SP-05-1209-PI-I-I. "Test of Nuclear Instrumentation Response to Diverse Logic System Actuations." Because the Main Steam Isolation Valves (MSIVs) were open at the time the Group I isolation signal was generated, they closed per design resulting in a reactor scram trip signal on MSIVs less than 90 percent open with the mode switch in the run position. Since the unit was in the cold shutdown condition at the time of the event, all controls rods were already inserted fully and therefore no rod motion occurred. However, the Main Steam Line Drain and Reactor Water Sample Line isolation valves also closed per design on the Group I isolation signal. Operations personnel returned the reactor mode switch to the shutdown position and reset the isolation and scram signals. Operations personnel left the MSIVs in the closed position. This event was caused by a personnel error resulting in the development of an inadequate procedure. The cause was that the test procedure was developed without the necessary steps to defeat the lower condenser vacuum isolation logic. The event in May 2016 was due to human performance of not following a procedure as written; therefore, the corrective actions from the 2009 event would not be expected to have prevented the condition reported in this LER.